第一章 原子的位形

库仑散射公式

$$b = \frac{a}{2}\cot\frac{\theta}{2} \qquad b \longleftrightarrow \theta$$

$$a \equiv \frac{1}{4\pi\varepsilon_0}\frac{Z_1 Z_2 e^2}{E}$$

卢瑟福公式

$$dp(\theta) = \frac{a^2 d\Omega}{16 A \sin^4\frac{\theta}{2}} n A t \cdot \qquad \left(a \equiv \frac{1}{4\pi\varepsilon_0}\frac{Z_1 Z_2 e^2}{E}\right)$$

$$= \frac{a^2 d\Omega}{16\sin^4\frac{\theta}{2}} nt \cdot \qquad = 2\pi nt\, b|db| \cdot \left(\frac{a^2 d\Omega}{16\sin^4\frac{\theta}{2}} = 2\pi b|db|\right).$$

① ② 单位面积金箔原子数.

$b = \frac{a}{2}\cot\frac{\theta}{2}$.

N个α粒子射出，dΩ方向上测得α粒子数应为：

$$dN' = \frac{Na^2 d\Omega}{16\sin^4\frac{\theta}{2}} nt = 2\pi N nt\, b|db|.$$

① ②

$$\Delta N = \int dN' = \int 2\pi N nt\, b|db| = \pi N nt\left|b_2^2 - b_1^2\right|$$

散射至 θ_1 到 θ_2 角度概率为

$$\frac{\Delta N}{N} = \pi nt\left|b_2^2 - b_1^2\right| = \pi nt\left(\frac{a}{2}\right)^2\left|\cot^2\frac{\theta_1}{2} - \cot^2\frac{\theta_2}{2}\right|$$

第二章. 原子的量子态.

光电效应.

$$\frac{1}{2}mv_m^2 = h\nu - \phi .$$

$$\nu_m = \frac{\phi}{h} \qquad \lambda = \frac{c}{\nu_m} = \frac{ch}{\phi}.$$

玻尔模型

H原子.

$$\frac{v^2}{r} m_e = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}$$

角动量量子化

$L = Vm_e r$

$= n\hbar$

$$\left.\begin{aligned} V^2 m_e^2 r^2 &= n^2\hbar^2 \\ V^2 m_e r &= \frac{1}{4\pi\varepsilon_0} e^2 \end{aligned}\right\} \quad r = \frac{n^2\hbar^2}{e^2 m_e}\cdot 4\pi\varepsilon_0$$

半数力学.

$$E_k = \frac{1}{2}m_e V^2 =$$

$$E_n = E_k + E_V = E_k - \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r} = -\frac{1}{2}\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

$$= -\frac{1}{2}\cdot\left(\frac{1}{4\pi\varepsilon_0}\right)^2 \frac{e^4 m_e}{n^2\hbar^2}$$

$$E_\infty = -E_1 = \frac{1}{2}m_e(\alpha c)^2 = Rhc = 13.6\ eV$$

类氢光谱.

将 e^2 换为 Ze^2.

第三章. 量子力学导论.

光的波粒二象性: $E = h\nu = p^2c^2 + E_0^2$ $(E_0 = 0)$.

$p = \frac{h}{\lambda} = \frac{E}{c}$

德布罗意公式. $\lambda = \frac{h}{p} = \frac{h}{mv}$.

质能方程: $E = E_k + E_0 = mc^2 = m_0c^2(1 + \frac{v^2}{c^2})$.

$(m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}})$.

相对论修正: $E^2 = p^2c^2 + m_0^2c^4$.

$\hbar c \approx 197\ MeV \cdot fm$.

不确定关系. $\Delta t \Delta E \geq \frac{\hbar}{2}$.

$\Delta x \Delta p_x \geq \frac{\hbar}{2}$

$E = p^2c^2 + m_0v^2$

第四章. 原子的精细结构，电子的自旋.

1. 磁矩.

$I = \frac{eV}{2\pi r}$

$$\vec{\mu} = IS\vec{n} = \frac{eV}{2\pi r}S\vec{n} = \frac{e}{m}mVS\vec{n} = -\frac{e}{2m_e}\vec{L} = -\gamma L$$

2. 原子态表达式.

$$^{2S+1}L_j$$

S , $j = L \pm S$. , L

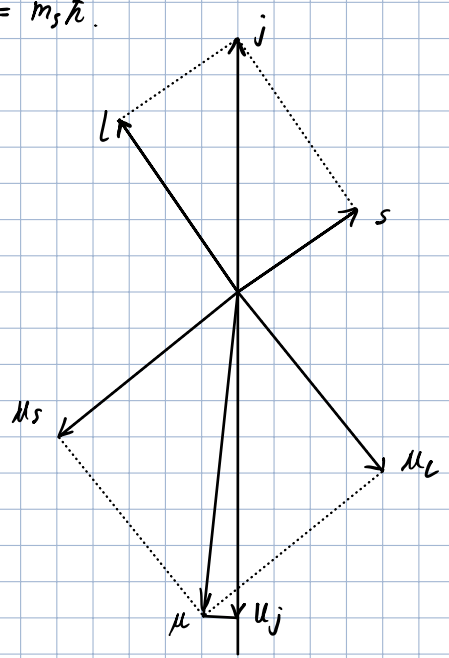
$m_j = -j, -(j-1), \quad j-1, \quad -[-(j-1)-1], \cdots\cdots$

$m_L = 0, \pm 1, \pm 2, \cdots, \pm L$

$m_S = \pm \frac{1}{2}$.

$m_j = m_L + m_S$.

$J_z = m_j\hbar$. $L_z = m_L\hbar$ $S_z = m_S\hbar$.



$$g = 1 + \frac{j(j+1) - L(L+1) + S(S+1)}{2j(j+1)}.$$

$$g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2j(j+1)}$$

$g_L = 1$ $g_S = 2$.

$\mu_L = -\sqrt{L(L+1)}\hbar g_L \mu_B = \frac{-e}{2m_e} g_L L$

$\mu_S = -\sqrt{S(S+1)}\hbar g_S \mu_B = \frac{-e}{2m_e} g_S S$.

$\mu_j = -\hbar\sqrt{j(j+1)} g_j \mu_B = \frac{-e}{2m_e} g_j j$.

3. 轨道自旋相互作用，能级分裂为2条.

能量差

$$\Delta U = \frac{(\alpha Z)^4}{n^3 l(l+1)} E_0.$$

$$= \frac{Z^4}{n^3 l(l+1)} \times 7.25 \times 10^{-4} eV$$

自旋磁矩在磁场中的能量. ⟶ 磁场强度.

$$U = -\mu_s \cdot B = -\mu_{sz} B = g_s m_s \mu_B B = \pm \mu_B B.$$

$$\Delta U = 2\mu_B B \implies B = \frac{\Delta U}{2\mu_B}.$$

第5章.

计算量子态 $^{2S+1}D_j$

S $\frac{3}{2}$ j $\frac{3}{2}$ L = 3.

$S = \sqrt{S(S+1)}\,\hbar$ $L = \sqrt{L(L+1)}\,\hbar$.

$J = \sqrt{J(J+1)}\,\hbar$

L-S 耦合.

$\vec{L} = \vec{L}_1 + \vec{L}_2$ $\vec{S} = \vec{S}_1 + \vec{S}_2$

$\vec{J} = \vec{L} + \vec{S}$ $J^2 = L^2 + S^2 + 2\vec{L}\cdot\vec{S}$.

$\vec{L}\cdot\vec{S} = \frac{1}{2}(J^2 - L^2 - S^2)$

$J = L+S,\ L+S-1,\ \cdots,\ |L-S|$ ← 已知 L, S, J 可能值.

$L = L_1+L_2,\ L_1+L_2-1,\ \cdots,\ |L_1-L_2|$

J-J.

$\vec{J}_1 = \vec{L}_1 + \vec{S}_1$ $\vec{J}_2 = \vec{L}_2 + \vec{S}_2$ $\vec{J} = \vec{J}_1 + \vec{J}_2$.